SUBRAHMANYESWAR KOLLURU

AI Engineer | Machine Learning Engineer | Computer Vision | LLM Systems
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PROFESSIONAL SUMMARY

Published AI researcher and B.Tech (AI and ML) undergraduate with hands-on experience building production-grade AI systems across computer vision, LLM applications, edge inference, and AI automation. Designed and deployed a TensorRT-optimized edge AI pipeline achieving 447 FPS at 2.23ms latency on standard CPU hardware without GPU. Published at ICCCNT 2025, IIT Indore. Skilled in multi-agent orchestration, retrieval-augmented generation, fine-tuning, and full-stack AI deployment. Seeking AI Engineer and Machine Learning Engineer roles for 2026 to 2027.

TECHNICAL SKILLS

Languages: Python, C++ (Basics), SQL
AI and ML: TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, YOLO, ArcFace, MobileNetV2, LSTM, Transfer Learning, Transformers
LLM and GenAI: LangChain, LlamaIndex, Ollama, Llama 3, OpenAI Whisper, RAG Pipelines, Vector Databases, Embeddings, LoRA, PEFT, HuggingFace, Fine-Tuning
AI Agents: CrewAI, Multi-Agent Systems, MCP Protocol, Agent Orchestration, Tool Integration
Deployment: FastAPI, TensorRT, Intel OpenVINO, FP16 Quantization, Edge AI, Docker, Streamlit, Next.js, Redis, WebSocket
Cloud: AWS S3, AWS EC2, AWS RDS, AWS Lambda, AWS IAM, MongoDB Atlas
Data and DB: MongoDB, Neo4j, MySQL, NumPy, Pandas, Matplotlib
CS Concepts: Data Structures and Algorithms (Intermediate), Object-Oriented Programming
Dev Tools: VS Code, Jupyter Notebook, Git, GitHub, Linux

WORK EXPERIENCE

AI Engineering Intern | READ Automation Pvt. Ltd., Chennai *Nov 2025 - Jan 2026*
Project: github.com/Subrahmanyeswar/Bottle-Cap-Inspection-System

- Engineered a 4-stage Edge AI inspection pipeline (Video Acquisition, Motion Trigger, ROI Extraction, MobileNetV2 Inference) achieving 447 FPS at 2.23ms latency on a standard laptop CPU, which is 23.8% faster than the TensorFlow Lite baseline of 361 FPS.
- Applied transfer learning on MobileNetV2 using ImageNet pre-trained weights with approximately 200 custom images and data augmentation, achieving 95% overall classification accuracy and 100% recall on critical Missing Cap defects.
- Optimized model inference with Intel OpenVINO FP16 quantization reducing model size by 48% from 8.9 MB to 4.6 MB and CPU utilization by 15.1%. Implemented motion-based triggering cutting idle CPU overhead by 60%. Applied CLAHE preprocessing ensuring accuracy above 92% across 200 to 2000 lux lighting. Research paper published.

PROJECTS

Aeternum Hub - GPU-Accelerated AI Security System | *Python, YOLO11m, TensorRT, ArcFace, ViT-GPT2, FastAPI, Redis, Next.js*
GitHub: github.com/Subrahmanyeswar/aeternum-hub

- Architected a full-stack GPU-accelerated AI security platform using YOLO11m with TensorRT achieving under 100ms detection latency at 25 to 30 FPS with 3 to 5x GPU speedup. Integrated ArcFace face recognition achieving 95% or higher accuracy with under 2% false positive rate using 512-dimensional embeddings.
- Built a 3-stage progressive alert system at 5, 10, and 20 seconds with automated SMS, voice calls, and video recording. Implemented ViT-GPT2 vision-language pipeline for AI-generated police reports. Designed 100% local processing architecture with zero cloud dependency, DPDP Act 2023 compliant. Published at ICCCNT 2025, IIT Indore. Top 10 National Project Expo. 3rd Prize national AI Hackathon.

Multi-Agent Competitive Intelligence Platform | *CrewAI, Python, LLMs, Multi-Agent Systems*
GitHub: github.com/Subrahmanyeswar/multi-agent-competitive-intelligence

- Designed a CrewAI multi-agent orchestration pipeline with three specialized agents (Research, Analysis, Synthesizer) for autonomous competitive landscape monitoring with parallel information processing and structured weekly intelligence report generation via agent-to-agent task delegation.

Advanced RAG System and LLM Infrastructure | *LangChain, LlamaIndex, Vector Database, Ollama, HuggingFace, LoRA, PEFT*
GitHub: github.com/Subrahmanyeswar/Aeternum-AI-Advanced-RAG

- Built an end-to-end retrieval-augmented generation pipeline with document ingestion, chunking, embedding generation, and vector similarity search for context-aware grounded LLM responses. Deployed local LLM inference via Ollama for offline zero-cost inference. Conducted LoRA and PEFT fine-tuning experiments on open-source LLMs using HuggingFace.

Jarvis AI Assistant | *Python, Groq API, Meta LLaMA 3, YOLOv8, OpenCV, Speech-to-Text, TTS*
GitHub: github.com/Subrahmanyeswar/Jarvis-AI-Assistant

- Built a modular voice-controlled AI assistant using Groq API with Meta LLaMA 3 for natural language processing, speech-to-text and TTS for hands-free interaction, and YOLOv8 with face recognition for real-time intruder detection. Engineered full PC automation via voice across 5 independent modules covering ai_core, productivity, pc_automation, system, and models.

EDUCATION

B.Tech - Computer Science and Engineering (AI and ML) *CGPA: 8.27 | 2023 - 2027*
Kalasalingam Academy of Research and Education (KARE), Virudhunagar, Tamil Nadu, India

ACHIEVEMENTS AND CERTIFICATIONS

- Research Publication - Published and presented at the 16th ICCCNT 2025, IIT Indore on AI security systems.
- National Project Expo - Top 10 finalist among 100 or more teams in a national-level engineering project competition.
- Hackathon - Awarded 3rd Prize in a national-level Artificial Intelligence and Machine Learning hackathon.